

TI C2000 Embedded Firmware: Closed-Loop SPWM Control for 500W SiC SEPIC Microinverter

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1 Control Strategy & Duty Cycle Derivation

The single-stage SEPIC inverter operating under Sinusoidal Pulse Width Modulation (SPWM) follows the complete static converter gain equation:

$$\frac{V_o}{V_{dc}} = \frac{D}{(1-D)^2} \quad (1)$$

Defining the normalized modulation index $M(t) = \frac{|v_{ref}(t)|}{V_{dc}}$, the instantaneous duty cycle $D(t)$ is explicitly solved via the quadratic formulation:

$$D(t) = \frac{1 + 2M - \sqrt{1 + 4M}}{2M} \quad (2)$$

2 Switch Logic Mapping

The firmware generates gate pulses across five SiC switches (S_1, S_2, S_3, S_4, S_5) across positive and negative grid half-cycles with configured dead-time limits ($t_d = 300$ ns, $t_{d,S1} = 200$ ns):

Table 1: Switch Operational States

Grid Half-Cycle	State	Active Switches
Positive ($v_{ref} > 0$)	Charging ($t_{sw} < T_{on}$)	$S_2 + S_4$
Positive ($v_{ref} > 0$)	Freewheeling ($t_{sw} > T_{on}$)	$S_4 + S_5$
Negative ($v_{ref} < 0$)	Charging ($t_{sw} < T_{on}$)	$S_1 + S_3 + S_5$
Negative ($v_{ref} < 0$)	Freewheeling ($t_{sw} > T_{on}$)	$S_4 + S_5$

3 TI C2000 Firmware Implementation

The following C routine is targeted for TI C2000 TMS320F28379D / F280049C microcontrollers:

```
1 #include <math.h>
2 #include <stdint.h>
3
4 #define FS          50000.0f
5 #define TS          (1.0f / FS)
6 #define TD          300e-9f
7 #define TD_S1       200e-9f
8 #define FO          50.0f
9 #define VO_RMS       200.0f
10 #define VDC         100.0f
11
```

```

12 #define TWO_PI      6.283185307179586f
13 #define SQRT_2      1.414213562373095f
14
15 typedef struct {
16     uint8_t S1, S2, S3, S4, S5;
17     float D;
18 } GatePulses_t;
19
20 GatePulses_t calculate_gate_pulses(float t) {
21     GatePulses_t pulses = {0, 0, 0, 0, 0, 0.0f};
22     float v_ref = (VO_RMS * SQRT_2) * sinf(TWO_PI * FO * t);
23
24     float M = fabsf(v_ref) / VDC;
25     float D = (M >= 1e-12f) ? ((1.0f + 2.0f * M - sqrtf(1.0f + 4.0f * M)) / (2.0f * M)
26 ) : 0.0f;
27
28     if (D < 0.0f) D = 0.0f;
29     if (D > 0.999f) D = 0.999f;
30     pulses.D = D;
31
32     float T_on = D * TS;
33     float t_sw = fmodf(t, TS);
34
35     if (v_ref >= 1e-3f) {
36         pulses.S4 = 1;
37         if ((t_sw > TD) && (t_sw < T_on)) {
38             pulses.S2 = 1;
39         } else if (t_sw > (T_on + TD)) {
40             pulses.S5 = 1;
41         }
42     } else if (v_ref <= -1e-3f) {
43         pulses.S5 = 1;
44         if ((t_sw > TD) && (t_sw < T_on)) {
45             pulses.S3 = 1;
46             if ((t_sw > (TD + TD_S1)) && (t_sw < T_on)) {
47                 pulses.S1 = 1;
48             } else {
49                 pulses.S5 = 0;
50             }
51         } else if (t_sw > (T_on + TD)) {
52             pulses.S4 = 1;
53         }
54     }
55     return pulses;
56 }

```

Listing 1: TI C2000 Gate Pulse Generation Firmware